

Gebze Technical University

Department of Electronics Engineering

ELM367: Digital Signal Processing

Lecture 5: Discrete Time Fourier Transform (DTFT)

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Fourier Transform Representation of Sequences

Discrete-Time Fourier Transform (DTFT)

$$H(e^{j\omega}) = \sum_{n=-\infty}^{\infty} h[n]e^{-j\omega n}$$

↓

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

↓

$X(e^{j\omega}) \rightarrow$ Continuous in frequency (ω) domain

$x[n] \rightarrow$ Discrete in time domain

Inverse DTFT (IDTFT):

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

- ▶ $X(e^{j\omega})$ is complex-valued function of ω and can be written in Cartesian or Polar forms:

$$X(e^{j\omega}) = X_R(e^{j\omega}) + jX_I(e^{j\omega}) = |X(e^{j\omega})| e^{j\angle X(e^{j\omega})}$$

Fourier Transform Representation of Sequences

Existence of $X(e^{j\omega})$

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$$

- ▶ This sum might not always converge.
- ▶ It only converges when $x[k]$ is absolutely summable:

$$\sum_{n=-\infty}^{\infty} |x[k]| < \infty$$

- ▶ This is the sufficient condition for the existence of $X(e^{j\omega})$

Fourier Transform Representation of Sequences

Existence of $X(e^{j\omega})$

Example: Find DTFT of $x[n] = a^n u[n]$

Fourier Transform Representation of Sequences

Special Cases: Fourier Transform of a Constant

Some signals are not absolutely summable: Example: $x[n] = 1$

$$\sum_{n=-\infty}^{\infty} |x[n]| = \sum_{n=-\infty}^{\infty} 1 = \cdots + 1 + 1 + 1 + 1 + \cdots = \infty$$

How can we find $X(e^{j\omega})$ for such systems?

Note that a constant means **zero frequency**, i.e., only exists when $\omega = 0$ for $-\pi \leq \omega \leq \pi$, therefore, we can write:

$$X(e^{j\omega}) = 2\pi \sum_{r=-\infty}^{\infty} \delta(\omega + 2\pi r)$$

For $x[n] = A$, we have

$$X(e^{j\omega}) = 2\pi A \sum_{r=-\infty}^{\infty} \delta(\omega + 2\pi r)$$

Fourier Transform Representation of Sequences

Special Cases: Fourier Transform of a Complex Exponential Coefficient

Assume that $X(e^{j\omega}) = 2\pi \sum_{r=-\infty}^{\infty} \delta(\omega - \omega_o + 2\pi r)$, what is $x[n]$?

By using Inverse DTFT

$$x[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} 2\pi \delta(\omega - \omega_o) e^{j\omega n} d\omega = e^{j\omega_o n}$$

Therefore, for

$$e^{j\omega_o n} \xLeftrightarrow{\text{DTFT}} 2\pi \sum_{r=-\infty}^{\infty} \delta(\omega - \omega_o + 2\pi r)$$

Symmetry Property Of The Fourier Transform

Conjugate Symmetric Sequences

Complex Conjugate

Example		Its Conjugate
$z = a + jb$	$\implies$	$z^* = a - jb$
$z = r \cdot e^{j\theta}$	$\implies$	$z^* = r \cdot e^{-j\theta}$

Conjugate Symmetric Sequences $x_e[n]$

$$x_e[n] = x_e^*[-n] \qquad X_e(e^{j\omega}) = X_e^*(e^{-j\omega})$$

Conjugate Anti-Symmetric Sequences $x_o[n]$

$$x_o[n] = -x_o^*[-n] \qquad X_o(e^{j\omega}) = -X_o^*(e^{-j\omega})$$

Symmetry Property Of The Fourier Transform

Conjugate Symmetric Sequences

Any sequence $x[n]$ can be expressed as a sum of conjugate symmetric $x_e[n]$ and conjugate anti-symmetric $x_o[n]$ sequences as:

$$x[n] = x_e[n] + x_o[n]$$

Proof: For a given sequence $x[n]$, let's construct $x_e[n]$ as

$$x_e[n] = \frac{1}{2} (x[n] + x^*[-n]) \quad \implies \text{This sequence is Conjugate symmetric}$$

$$x_e^*[n] = \frac{1}{2} (x[n] + x^*[-n])^*$$

$$x_e^*[n] = \frac{1}{2} (x^*[n] + x[-n])$$

$$x_e^*[-n] = \frac{1}{2} (x^*[-n] + x[n])$$

$$x_e^*[-n] = x_e[n]$$

Symmetry Property Of The Fourier Transform

Conjugate Symmetric Sequences

Similarly, let's construct $x_o[n]$ as

$$x_o[n] = \frac{1}{2} (x[n] - x^*[-n]) \implies \text{This sequence is Conjugate anti-symmetric}$$

$$x_o^*[n] = \frac{1}{2} (x[n] - x^*[-n])^*$$

$$x_o^*[n] = \frac{1}{2} (x^*[n] - x[-n])$$

$$x_o^*[-n] = \frac{1}{2} (x^*[-n] - x[n])$$

$$x_o^*[-n] = -\frac{1}{2} (x[n] - x^*[-n])$$

$$x_o^*[-n] = -x_o[n] \implies x_o[n] = -x_o^*[-n]$$

Symmetry Property Of The Fourier Transform

Conjugate Symmetric Sequences

Therefore, we can have:

$$\begin{array}{rcl} & x_e[n] = \frac{1}{2} (x[n] + x^*[-n]) \\ + & x_o[n] = \frac{1}{2} (x[n] - x^*[-n]) \\ \hline x_e[n] + x_o[n] = & & x[n] \end{array}$$

Even and Odd Sequences

- ▶ A **real sequence** that is **conjugate symmetric**, i.e., $x_e[n] = x_e^*[-n]$, is referred to as an **Even Sequence**
- ▶ A **real sequence** that is **conjugate anti-symmetric**, i.e., $x_o[n] = -x_o^*[-n]$, is referred to as an **Odd Sequence**

Symmetry Property Of The Fourier Transform

Conjugate Symmetric Sequences

DTFT Domain

Any Fourier transform $X(e^{j\omega})$ can be decomposed into a sum of conjugate symmetric $X_e(e^{j\omega})$ and conjugate anti-symmetric $X_o(e^{j\omega})$ functions as

$$X(e^{j\omega}) = X_e(e^{j\omega}) + X_o(e^{j\omega})$$

Where

$$X_e(e^{j\omega}) = \frac{1}{2} [X(e^{j\omega}) + X^*(e^{-j\omega})] \quad \implies \text{Conjugate symmetric}$$

and

$$X_o(e^{j\omega}) = \frac{1}{2} [X(e^{j\omega}) - X^*(e^{-j\omega})] \quad \implies \text{Conjugate anti-symmetric}$$

Symmetry Property Of The Fourier Transform

Conjugate Symmetric Sequences

TABLE 2.1 SYMMETRY PROPERTIES OF THE FOURIER TRANSFORM

Sequence $x[n]$	Fourier Transform $X(e^{j\omega})$
1. $x^*[n]$	$X^*(e^{-j\omega})$
2. $x^*[-n]$	$X^*(e^{j\omega})$
3. $\mathcal{R}\{x[n]\}$	$X_e(e^{j\omega})$ (conjugate-symmetric part of $X(e^{j\omega})$)
4. $j\mathcal{I}\{x[n]\}$	$X_o(e^{j\omega})$ (conjugate-antisymmetric part of $X(e^{j\omega})$)
5. $x_e[n]$ (conjugate-symmetric part of $x[n]$)	$X_R(e^{j\omega}) = \mathcal{R}\{X(e^{j\omega})\}$
6. $x_o[n]$ (conjugate-antisymmetric part of $x[n]$)	$jX_I(e^{j\omega}) = j\mathcal{I}\{X(e^{j\omega})\}$
<i>The following properties apply only when $x[n]$ is real:</i>	
7. Any real $x[n]$	$X(e^{j\omega}) = X^*(e^{-j\omega})$ (Fourier transform is conjugate symmetric)
8. Any real $x[n]$	$X_R(e^{j\omega}) = X_R(e^{-j\omega})$ (real part is even)
9. Any real $x[n]$	$X_I(e^{j\omega}) = -X_I(e^{-j\omega})$ (imaginary part is odd)
10. Any real $x[n]$	$ X(e^{j\omega}) = X(e^{-j\omega}) $ (magnitude is even)
11. Any real $x[n]$	$\angle X(e^{j\omega}) = -\angle X(e^{-j\omega})$ (phase is odd)
12. $x_e[n]$ (even part of $x[n]$)	$X_R(e^{j\omega})$
13. $x_o[n]$ (odd part of $x[n]$)	$jX_I(e^{j\omega})$

Symmetry Property Of The Fourier Transform

Illustration of the Symmetry Properties

Example: Given a real sequence $x[n] = a^n u[n]$ where $|a| < 1$, whose Fourier transform is $X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}}$

$$X(e^{j\omega}) = \frac{1}{1 - ae^{-j\omega}} = X^*(e^{-j\omega}) \quad (\text{property 7}),$$

$$X_R(e^{j\omega}) = \frac{1 - a \cos \omega}{1 + a^2 - 2a \cos \omega} = X_R(e^{-j\omega}) \quad (\text{property 8}),$$

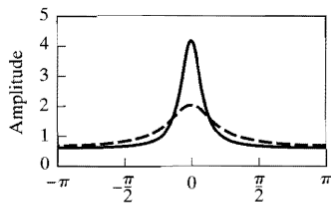
$$X_I(e^{j\omega}) = \frac{-a \sin \omega}{1 + a^2 - 2a \cos \omega} = -X_I(e^{-j\omega}) \quad (\text{property 9}),$$

$$|X(e^{j\omega})| = \frac{1}{(1 + a^2 - 2a \cos \omega)^{1/2}} = |X(e^{-j\omega})| \quad (\text{property 10}),$$

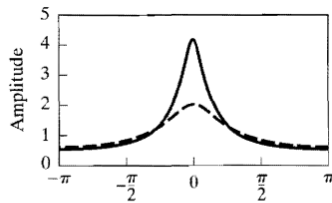
$$\angle X(e^{j\omega}) = \tan^{-1} \left(\frac{-a \sin \omega}{1 - a \cos \omega} \right) = -\angle X(e^{-j\omega}) \quad (\text{property 11}).$$

Symmetry Property Of The Fourier Transform

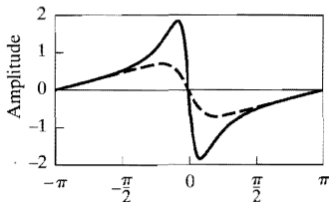
Illustration of the Symmetry Properties



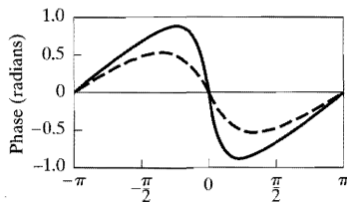
(a)



(c)



(b)



(d)

(a) Real and Imaginary parts

(b) Magnitude and Phase

Symmetry Property Of The Fourier Transform

Conjugate Symmetric Sequences

Why are we interested in Symmetry Properties?

- ▶ Reduces computational complexity:
 - ▶ We can only compute half of the transform and use the symmetry property to get the other half. Example: Fast Fourier Transform (FFT) algorithm
- ▶ Gives insight into how the signal behaves in the other domain and simplifies the inverse Fourier transform process.
 - ▶ Example: if $X(e^{j\omega}) = X^*(e^{-j\omega})$ we know that the time domain signal is real-valued.

Fourier Transform Theorems

Notation

DTFT Pairs

$$x[n] \quad \stackrel{\mathcal{F}}{\longleftrightarrow} \quad X(e^{j\omega})$$

DTFT

$$\begin{aligned} X(e^{j\omega}) &= \mathcal{F}\{x[n]\} \\ &= \sum_{n=-\infty}^{\infty} x[n] e^{-j\omega n} \end{aligned}$$

Inverse DTFT

$$\begin{aligned} x[n] &= \mathcal{F}^{-1}\{X(e^{j\omega})\} \\ &= \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega \end{aligned}$$

Fourier Transform Theorems

TABLE 2.2 FOURIER TRANSFORM THEOREMS

Sequence	Fourier Transform
$x[n]$	$X(e^{j\omega})$
$y[n]$	$Y(e^{j\omega})$
1. $ax[n] + by[n]$	$aX(e^{j\omega}) + bY(e^{j\omega})$
2. $x[n - n_d]$ (n_d an integer)	$e^{-j\omega n_d} X(e^{j\omega})$
3. $e^{j\omega_0 n} x[n]$	$X(e^{j(\omega - \omega_0)})$
4. $x[-n]$	$X(e^{-j\omega})$ $X^*(e^{j\omega})$ if $x[n]$ real.
5. $nx[n]$	$j \frac{dX(e^{j\omega})}{d\omega}$
6. $x[n] * y[n]$	$X(e^{j\omega})Y(e^{j\omega})$
7. $x[n]y[n]$	$\frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\theta})Y(e^{j(\omega - \theta)})d\theta$

Parseval's theorem:

$$8. \sum_{n=-\infty}^{\infty} |x[n]|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

$$9. \sum_{n=-\infty}^{\infty} x[n]y^*[n] = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega})Y^*(e^{j\omega})d\omega$$

Fourier Transform Theorems

TABLE 2.3 FOURIER TRANSFORM PAIRS

Sequence	Fourier Transform
1. $\delta[n]$	1
2. $\delta[n - n_0]$	$e^{-j\omega n_0}$
3. 1 $(-\infty < n < \infty)$	$\sum_{k=-\infty}^{\infty} 2\pi \delta(\omega + 2\pi k)$
4. $a^n u[n]$ $(a < 1)$	$\frac{1}{1 - ae^{-j\omega}}$
5. $u[n]$	$\frac{1}{1 - e^{-j\omega}} + \sum_{k=-\infty}^{\infty} \pi \delta(\omega + 2\pi k)$
6. $(n+1)a^n u[n]$ $(a < 1)$	$\frac{1}{(1 - ae^{-j\omega})^2}$
7. $\frac{r^n \sin \omega_p (n+1)}{\sin \omega_p} u[n]$ $(r < 1)$	$\frac{1}{1 - 2r \cos \omega_p e^{-j\omega} + r^2 e^{-j2\omega}}$
8. $\frac{\sin \omega_c n}{\pi n}$	$X(e^{j\omega}) = \begin{cases} 1, & \omega < \omega_c, \\ 0, & \omega_c < \omega \leq \pi \end{cases}$
9. $x[n] = \begin{cases} 1, & 0 \leq n \leq M \\ 0, & \text{otherwise} \end{cases}$	$\frac{\sin[\omega(M+1)/2]}{\sin(\omega/2)} e^{-j\omega M/2}$
10. $e^{j\omega_0 n}$	$\sum_{k=-\infty}^{\infty} 2\pi \delta(\omega - \omega_0 + 2\pi k)$
11. $\cos(\omega_0 n + \phi)$	$\sum_{k=-\infty}^{\infty} [\pi e^{j\phi} \delta(\omega - \omega_0 + 2\pi k) + \pi e^{-j\phi} \delta(\omega + \omega_0 + 2\pi k)]$

Fourier Transform Theorems

Examples: Determining Fourier Transform Using the Table

Given a sequence $x[n] = a^n u[n - 5]$. Find $X(e^{j\omega})$

Fourier Transform Theorems

Examples: Determining Inverse Fourier Transform Using the Table

Given a sequence $X(e^{j\omega}) = \frac{1}{(1-ae^{-j\omega})(1-be^{-j\omega})}$. Find $x[n]$

Fourier Transform Theorems

Examples: Determining Impulse Response From a Frequency Response

Given a frequency response of a highpass filter $H_{\text{hp}}(e^{j\omega})$ as:

$$H_{\text{hp}}(e^{j\omega}) = \begin{cases} e^{-j\omega n_d} & , \omega_c < |\omega| < \pi \\ 0 & , |\omega| < \omega_c \end{cases}$$

Find $h_{\text{hp}}[n]$

Fourier Transform Theorems

Examples: Determining Impulse Response From a Difference Equation

Given a sequence $y[n] - \frac{1}{2}y[n-1] = x[n] - \frac{1}{4}x[n-1]$. Find $h[n]$